

Lattice-based cryptographic constructions

Elena Kirshanova

at Post-Quantum Cryptography Summer School
Hammamet, Tunisia

There will be two lectures on lattices:

1. Lattice-based cryptographic constructions (now)
2. Cryptanalysis of lattice-based constructions (tomorrow)

Lecture materials are available at
TODO: QR

Agenda

1. Learning with Errors (LWE)
2. Algebraic versions of LWE and Kyber
3. Short Integer Solution problem (SIS)
4. Dilithium Signature

Learning with Errors (LWE) [Regev'05]

Fix dimension $n > 0$, modulus $q > 1$, and noise distribution χ

Search LWE: Find secret $\mathbf{s} \in \mathbb{Z}_q^n$ given noisy inner products:

$$\begin{aligned} \mathbf{a}_1 &\stackrel{\$}{\leftarrow} \mathbb{Z}_q^n, \quad e_1 \leftarrow \chi, & (\mathbf{a}_1, \langle \mathbf{a}_1, \mathbf{s} \rangle + e_1 \bmod q), \\ & & \vdots \\ \mathbf{a}_m &\stackrel{\$}{\leftarrow} \mathbb{Z}_q^n, \quad e_m \leftarrow \chi, & (\mathbf{a}_m, \langle \mathbf{a}_m, \mathbf{s} \rangle + e_m \bmod q) \end{aligned}$$

Learning with Errors (LWE) [Regev'05]

Fix dimension $n > 0$, modulus $q > 1$, and noise distribution χ

Search LWE: Find secret $\mathbf{s} \in \mathbb{Z}_q^n$ given noisy inner products:

$$\begin{aligned} \mathbf{a}_1 &\stackrel{\$}{\leftarrow} \mathbb{Z}_q^n, e_1 \leftarrow \chi, & (\mathbf{a}_1, \langle \mathbf{a}_1, \mathbf{s} \rangle + e_1 \bmod q), \\ & & \vdots \\ \mathbf{a}_m &\stackrel{\$}{\leftarrow} \mathbb{Z}_q^n, e_m \leftarrow \chi, & (\mathbf{a}_m, \langle \mathbf{a}_m, \mathbf{s} \rangle + e_m \bmod q) \end{aligned}$$

Decision LWE: Distinguish between the two distributions:

$$\begin{aligned} \mathbf{a}_1 &\stackrel{\$}{\leftarrow} \mathbb{Z}_q^n, b \stackrel{\$}{\leftarrow} \mathbb{Z}_q \\ &\vdots \\ \mathbf{a}_m &\stackrel{\$}{\leftarrow} \mathbb{Z}_q^n, b \stackrel{\$}{\leftarrow} \mathbb{Z}_q \end{aligned}$$

$$\begin{aligned} \mathbf{a}_1 &\stackrel{\$}{\leftarrow} \mathbb{Z}_q^n, b = \langle \mathbf{a}_1, \mathbf{s} \rangle + e_1 \bmod q \\ &\vdots \\ \mathbf{a}_m &\stackrel{\$}{\leftarrow} \mathbb{Z}_q^n, b = \langle \mathbf{a}_m, \mathbf{s} \rangle + e_m \bmod q \end{aligned}$$

LWE: numerical example

$n = 1$, $q = 13$, χ is central binomial distribution over $\{-1, 0, 1\}$

$$\Pr[e = -1] = \Pr[e = 1] = \frac{1}{4}, \Pr[e = 0] = \frac{1}{2}.$$

Can you guess $\mathbf{s}$ from

$$\mathbf{a}_1 = (10), b_1 = 9$$

$$\mathbf{a}_2 = (5), b_2 = 5$$

LWE: numerical example

$n = 1$, $q = 13$, χ is central binomial distribution over $\{-1, 0, 1\}$

$$\Pr[e = -1] = \Pr[e = 1] = \frac{1}{4}, \Pr[e = 0] = \frac{1}{2}.$$

Can you guess **s** from

$$\mathbf{a}_1 = (10), b_1 = 9$$

$$\mathbf{a}_2 = (5), b_2 = 5$$

$$\mathbf{a}_3 = (8), b_3 = 9$$

$$\mathbf{a}_4 = (11), b_4 = 11$$

LWE: numerical example

$n = 1$, $q = 13$, χ is central binomial distribution over $\{-1, 0, 1\}$

$$\Pr[e = -1] = \Pr[e = 1] = \frac{1}{4}, \Pr[e = 0] = \frac{1}{2}.$$

Can you guess **s** from

$$\mathbf{a}_1 = (10), b_1 = 9$$

$$\mathbf{a}_2 = (5), b_2 = 5$$

$$\mathbf{a}_3 = (8), b_3 = 9$$

$$\mathbf{a}_4 = (11), b_4 = 11$$

$$\mathbf{a}_5 = (12), b_4 = 0$$

LWE: numerical example

$n = 1$, $q = 13$, χ is central binomial distribution over $\{-1, 0, 1\}$

$$\Pr[e = -1] = \Pr[e = 1] = \frac{1}{4}, \Pr[e = 0] = \frac{1}{2}.$$

Can you guess $\mathbf{s}$ from

$$\mathbf{a}_1 = (10), b_1 = 9$$

$$\mathbf{a}_2 = (5), b_2 = 5$$

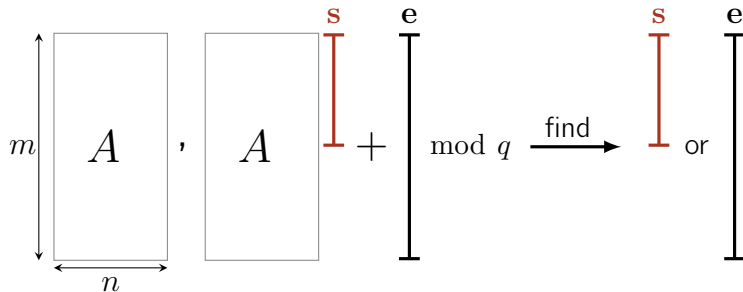
$$\mathbf{a}_3 = (8), b_3 = 9$$

$$\mathbf{a}_4 = (11), b_4 = 11$$

$$\mathbf{a}_5 = (12), b_5 = 0$$

$$\mathbf{s} = (1)$$

LWE in matrix form

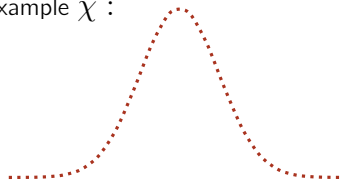


$$A \xleftarrow{\$} \mathbb{Z}_q^{m \times n}$$

$$\mathbf{s} \xleftarrow{\$} \mathbb{Z}_q^n$$

$$\mathbf{e} \xleftarrow{\$} \chi^m$$

Example χ :



LWE hardness

- ▶ The solution is unique for enough number of LWE samples $(\mathbf{a}, b)$ ($\#$ samples $> n \log q$ is enough).
- ▶ The larger n , the harder LWE. In crypto we use $n \geq 512$
- ▶ Modulus q is typically $\text{poly}(n)$, i.e., $q = n^2$
- ▶ Exact distribution of the secret does not play a significant role, as long as $\mathbf{s}$ has enough entropy. Binary or ternary $\mathbf{s}$ are common choices.
- ▶ The distribution of $\mathbf{e}$ is important: $\mathbf{e} = 0$ makes LWE trivially easy (why?), if $\mathbf{e} \xleftarrow{\$} \mathbb{Z}_q^m$ makes LWE trivially hard. Common choices: discrete Gaussian, central binomial, ternary.
- ▶ Best known attack on LWE runs in time $\sim 2^{\mathcal{O}(n)}$.

...more about LWE hardness tomorrow.

Search vs. Decision LWE

Decision-to-Search: Trivial

Search-to-Decision: Let $\mathbf{s}^*$ be the secret. Goal: find $\mathbf{s}_1^*$ with access to decision LWE oracle and with $\{(\mathbf{a}_i, b_i)\}$.

Search vs. Decision LWE

Decision-to-Search: Trivial

Search-to-Decision: Let $\mathbf{s}^*$ be the secret. Goal: find $\mathbf{s}_1^*$ with access to decision LWE oracle and with $\{(\mathbf{a}_i, b_i)\}$.

1. Make a guess $s'_1 \xleftarrow{\$} \mathbb{Z}_q$ for $\mathbf{s}_1^*$. Call Decision LWE oracle on modified samples

$$\mathbf{a}_i + (1, 0, \dots, 0)^T, b_i + s'_1$$

Search vs. Decision LWE

Decision-to-Search: Trivial

Search-to-Decision: Let $\mathbf{s}^*$ be the secret. Goal: find $\mathbf{s}_1^*$ with access to decision LWE oracle and with $\{(\mathbf{a}_i, b_i)\}$.

1. Make a guess $s'_1 \xleftarrow{\$} \mathbb{Z}_q$ for $\mathbf{s}_1^*$. Call Decision LWE oracle on modified samples

$$\mathbf{a}_i + (1, 0, \dots, 0)^T, b_i + s'_1$$

- If $s'_1 = \mathbf{s}_1^*$,

$$b_i + s'_1 = \langle \mathbf{a}_i, \mathbf{s} \rangle + e + s'_1 = \langle \mathbf{a}_i, \mathbf{s} \rangle + e + \langle (1, 0, \dots, 0), \mathbf{s} \rangle = \langle \mathbf{a}_i + (1, 0, \dots, 0), \mathbf{s} \rangle + e$$

Search vs. Decision LWE

Decision-to-Search: Trivial

Search-to-Decision: Let $\mathbf{s}^*$ be the secret. Goal: find $\mathbf{s}_1^*$ with access to decision LWE oracle and with $\{(\mathbf{a}_i, b_i)\}$.

1. Make a guess $s'_1 \xleftarrow{\$} \mathbb{Z}_q$ for $\mathbf{s}_1^*$. Call Decision LWE oracle on modified samples

$$\mathbf{a}_i + (1, 0, \dots, 0)^T, b_i + s'_1$$

► If $s'_1 = \mathbf{s}_1^*$, Correct LWE sample!
$$b_i + s'_1 = \langle \mathbf{a}_i, \mathbf{s} \rangle + e + s'_1 = \langle \mathbf{a}_i, \mathbf{s} \rangle + e + \langle (1, 0, \dots, 0), \mathbf{s} \rangle = \langle \mathbf{a}_i + (1, 0, \dots, 0), \mathbf{s} \rangle + e$$

► If $s'_1 \neq \mathbf{s}_1^*$,

$$b_i + s'_1 \text{ is uniform in } \mathbb{Z}_q.$$

$\implies$ LWE Decision oracle distinguishes correct guess from the incorrect ones.

PKE from LWE

Alice



$$\text{pk} = (A \in \mathbb{Z}_q^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathbb{Z}_q^m)$$
$$\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$$

Bob



$$\mu \in \{0, 1\}$$

PKE from LWE

Alice



$$\text{pk} = (A \in \mathbb{Z}_q^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathbb{Z}_q^m)$$
$$\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$$

Bob



$$\mu \in \{0, 1\}$$

Encryption:

$$\mathbf{t}_1 \leftarrow \chi^m, \mathbf{t}_2 \leftarrow \chi^n, \mathbf{t}_3 \leftarrow \chi^m$$
$$\mathbf{c}_1 = \mathbf{t}_1^T \cdot A + \mathbf{t}_2^T \bmod q$$

PKE from LWE

Alice



$$\text{pk} = (A \in \mathbb{Z}_q^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathbb{Z}_q^m)$$
$$\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$$

Bob



$$\mu \in \{0, 1\}$$

Encryption:

$$\mathbf{t}_1 \leftarrow \chi^m, \mathbf{t}_2 \leftarrow \chi^n, \mathbf{t}_3 \leftarrow \chi^m$$

$$\mathbf{c}_1 = \mathbf{t}_1^T \cdot A + \mathbf{t}_2^T \bmod q$$

$$\xleftarrow{c = (\mathbf{c}_1, c_2)} \quad c_2 = \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 + \mu \lfloor \frac{q}{2} \rfloor \bmod q$$

PKE from LWE

Alice



$$\text{pk} = (A \in \mathbb{Z}_q^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathbb{Z}_q^m)$$
$$\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$$

Decryption:

$$\text{if } |c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle| \text{ close to } 0 \implies \mu = 0$$

Bob



$$\mu \in \{0, 1\}$$

Encryption:

$$\mathbf{t}_1 \leftarrow \chi^m, \mathbf{t}_2 \leftarrow \chi^n, \mathbf{t}_3 \leftarrow \chi^m$$

$$\mathbf{c}_1 = \mathbf{t}_1^T \cdot A + \mathbf{t}_2^T \bmod q$$

$$\xleftarrow{c = (\mathbf{c}_1, c_2)} \quad c_2 = \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 + \mu \lfloor \frac{q}{2} \rfloor \bmod q$$

PKE from LWE

Alice



$$\text{pk} = (A \in \mathbb{Z}_q^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathbb{Z}_q^m)$$

$$\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$$

Decryption:

$$\text{if } |c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle| \text{ close to } 0 \implies \mu = 0$$

Correctness:

$$c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle =$$

$$\langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 - \mathbf{t}_1^T A \mathbf{s} - \mathbf{t}_2^T \mathbf{s} + \mu \lfloor \frac{q}{2} \rfloor$$

$$=$$

Bob



$$\mu \in \{0, 1\}$$

Encryption:

$$\mathbf{t}_1 \leftarrow \chi^m, \mathbf{t}_2 \leftarrow \chi^n, \mathbf{t}_3 \leftarrow \chi^m$$

$$\mathbf{c}_1 = \mathbf{t}_1^T \cdot A + \mathbf{t}_2^T \bmod q$$

$$\xleftarrow{c = (\mathbf{c}_1, c_2)} \quad c_2 = \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 + \mu \lfloor \frac{q}{2} \rfloor \bmod q$$

PKE from LWE

Alice



pk = $(A \in \mathbb{Z}_q^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathbb{Z}_q^m)$
 $\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$

Correctness:

$$\begin{aligned} c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle &= \\ \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 - \mathbf{t}_1^T A \mathbf{s} - \mathbf{t}_2^T \mathbf{s} + \mu \lfloor \frac{q}{2} \rfloor &= \\ \mathbf{t}_1^T A \mathbf{s} + \mathbf{t}_1^T \mathbf{e} + \mathbf{t}_3 - \mathbf{t}_1^T A \mathbf{s} - \mathbf{t}_2^T \mathbf{s} + \mu \lfloor \frac{q}{2} \rfloor &= \\ \mathbf{t}_1^T \mathbf{e} + \mathbf{t}_3 - \mathbf{t}_2^T \mathbf{s} + \mu \lfloor \frac{q}{2} \rfloor &= \end{aligned}$$

Bob



$\mu \in \{0, 1\}$

Encryption:

$\mathbf{t}_1 \leftarrow \chi^m, \mathbf{t}_2 \leftarrow \chi^n, \mathbf{t}_3 \leftarrow \chi^m$

$\mathbf{c}_1 = \mathbf{t}_1^T \cdot A + \mathbf{t}_2^T \bmod q$

$\xleftarrow{c=(\mathbf{c}_1, c_2)} c_2 = \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 + \mu \lfloor \frac{q}{2} \rfloor \bmod q$

Decryption:

if $|c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle|$ close to 0 $\implies \mu = 0$

PKE from LWE

Alice



pk = $(A \in \mathbb{Z}_q^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathbb{Z}_q^m)$
 $\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$

Correctness:

$$\begin{aligned} c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle &= \\ \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 - \mathbf{t}_1^T A \mathbf{s} - \mathbf{t}_2^T \mathbf{s} + \mu \lfloor \frac{q}{2} \rfloor &= \\ \mathbf{t}_1^T A \mathbf{s} + \mathbf{t}_1^T \mathbf{e} + \mathbf{t}_3 - \mathbf{t}_1^T A \mathbf{s} - \mathbf{t}_2^T \mathbf{s} + \mu \lfloor \frac{q}{2} \rfloor &= \\ \underbrace{\mathbf{t}_1^T \mathbf{e} + \mathbf{t}_3 - \mathbf{t}_2^T \mathbf{s}}_{\text{small}} + \mu \lfloor \frac{q}{2} \rfloor \end{aligned}$$

Bob



$\mu \in \{0, 1\}$

Encryption:

$\mathbf{t}_1 \leftarrow \chi^m, \mathbf{t}_2 \leftarrow \chi^n, \mathbf{t}_3 \leftarrow \chi^m$

$\mathbf{c}_1 = \mathbf{t}_1^T \cdot A + \mathbf{t}_2^T \bmod q$

$\xleftarrow{c = (\mathbf{c}_1, c_2)} c_2 = \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 + \mu \lfloor \frac{q}{2} \rfloor \bmod q$

Decryption:

if $|c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle|$ close to 0 $\implies \mu = 0$

PKE from LWE. Security

Alice



$$\text{pk} = (A \in \mathbb{Z}_q^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathbb{Z}_q^m)$$
$$\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$$

Decryption:

$$\text{if } |c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle| \text{ close to } 0 \implies \mu = 0$$

Bob



$$\mu \in \{0, 1\}$$

Encryption:

$$\mathbf{t}_1 \leftarrow \chi^m, \mathbf{t}_2 \leftarrow \chi^n, \mathbf{t}_3 \leftarrow \chi^m$$

$$\mathbf{c}_1 = \mathbf{t}_1^T \cdot A + \mathbf{t}_2^T \bmod q$$

$$\xleftarrow{c = (\mathbf{c}_1, c_2)} \quad c_2 = \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 + \mu \lfloor \frac{q}{2} \rfloor \bmod q$$

PKE from LWE. Security

Alice



Indistinguishable under Decision LWE

Bob



$$\text{pk} = (A \in \mathbb{Z}_q^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathbb{Z}_q^m)$$

$$\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$$

$$\mu \in \{0, 1\}$$

Encryption:

$$\mathbf{t}_1 \leftarrow \chi^m, \mathbf{t}_2 \leftarrow \chi^n, \mathbf{t}_3 \leftarrow \chi^m$$

$$\mathbf{c}_1 = \mathbf{t}_1^T \cdot A + \mathbf{t}_2^T \bmod q$$

$$\xleftarrow{c = (\mathbf{c}_1, c_2)} c_2 = \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 + \mu \lfloor \frac{q}{2} \rfloor \bmod q$$

Decryption:

$$\text{if } |c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle| \text{ close to } 0 \implies \mu = 0$$

PKE from LWE. Security

Alice



Indistinguishable under Decision LWE

$$\text{pk} = (A \in \mathbb{Z}_q^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathbb{Z}_q^m)$$

$$\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$$

Bob



$$\mu \in \{0, 1\}$$

Indistinguishable under Decision LWE

Encryption:

$$\mathbf{t}_1 \leftarrow \chi^m, \mathbf{t}_2 \leftarrow \chi^n, \mathbf{t}_3 \leftarrow \chi^m$$

$$\mathbf{c}_1 = \mathbf{t}_1^T \cdot A + \mathbf{t}_2^T \bmod q$$

$$c_2 = \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 + \mu \lfloor \frac{q}{2} \rfloor \bmod q$$

$$c = (c_1, c_2)$$

Decryption:

$$\text{if } |c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle| \text{ close to } 0 \implies \mu = 0$$

Algebraic LWE: motivation

- ▶ Storing LWE samples requires $\Omega(n^2 \log q)$ bits
 - ▶ Matrix vector multiplication require $O(n^2)$ operations in $\mathbb{Z}_q$
- $\implies$ 'standard' LWE is inefficient¹

Solution: Algebraic versions of LWE

Options: polynomial LWE, Ring-LWE, Module-LWE, ...

¹Although concrete PKE proposals exist (e.g., Frodo) with pk and ct $\approx$ 10kB

Polynomial rings

- ▶ $\mathbb{Z}[x]$ – polynomial ring (i.e., polynomial with integer coefficients)
- ▶ $\mathbb{Z}_q[x]$ – quotient polynomial ring i.e., polynomials with coefficients taken mod q
- ▶ $f \in \mathbb{Z}[x]$ is **unitary** if its leading coefficient is 1 (example: $f = x^4 + 4x^3 + 7$)
- ▶ f is **irreducible** in $\mathbb{Z}[x]$ if it does not have roots in $\mathbb{Z}[x]$ (example: $f = x^2 + 1$)
- ▶ $\mathbb{Z}[x]/f$ polynomial quotient ring, i.e., set of polynomials taken mod f
- ▶ To multiply $g, h \in \mathbb{Z}[x]/f$, first compute $r = g \cdot h$ using ordinary polynomial multiplication, $\deg(r) \leq 2n - 2$ then divide r by f and take the remainder of the division.

Polynomial multiplication as matrix-vector product

Any polynomial can be represented as a vector via [coefficient embedding](#).

For $f \in \mathbb{Z}[x]$ a unitary irreducible of degree n :

$$a = \sum_i a_i x^i \in \mathbb{Z}[x]/f \longrightarrow (a_0, \dots, a_{n-1}) \in \mathbb{Z}^n$$

Polynomial multiplication as matrix-vector product

Any polynomial can be represented as a vector via [coefficient embedding](#).

For $f \in \mathbb{Z}[x]$ a unitary irreducible of degree n :

$$a = \sum_i a_i x^i \in \mathbb{Z}[x]/f \longrightarrow (a_0, \dots, a_{n-1}) \in \mathbb{Z}^n$$

For $a, b \in \mathbb{Z}[x]/f$ and $c = a \cdot b \in \mathbb{Z}[x]/f$, it holds

$$\begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-2} \\ c_{n-1} \end{bmatrix} = \begin{bmatrix} | & | & \dots & | \\ | & | & \dots & | \\ \text{coeff}(a) & \text{coeff}(x \cdot a) & \ddots & \text{coeff}(x^{n-1} \cdot a) \\ | & | & \dots & | \\ | & | & \dots & | \end{bmatrix} \cdot \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{n-2} \\ b_{n-1} \end{bmatrix}$$

Polynomial multiplication as matrix-vector product

Any polynomial can be represented as a vector via [coefficient embedding](#).

For $f \in \mathbb{Z}[x]$ a unitary irreducible of degree n :

$$a = \sum_i a_i x^i \in \mathbb{Z}[x]/f \longrightarrow (a_0, \dots, a_{n-1}) \in \mathbb{Z}^n$$

Example: $f = x^n + 1$:

$$\begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-2} \\ c_{n-1} \end{bmatrix} = \begin{bmatrix} a_0 & -a_{n-1} & \dots & -a_1 \\ a_1 & a_0 & \dots & -a_2 \\ \vdots & \vdots & \ddots & \\ a_{n-2} & a_{n-3} & \dots & -a_{n-1} \\ \underbrace{a_{n-1}}_a & \underbrace{a_{n-2}}_{x \cdot a} & \dots & \underbrace{a_0}_{x^{n-1} \cdot a} \end{bmatrix} \cdot \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{n-2} \\ b_{n-1} \end{bmatrix}$$

Polynomial multiplication as matrix-vector product

Any polynomial can be represented as a vector via [coefficient embedding](#).

For $f \in \mathbb{Z}[x]$ a unitary irreducible of degree n :

$$a = \sum_i a_i x^i \in \mathbb{Z}[x]/f \longrightarrow (a_0, \dots, a_{n-1}) \in \mathbb{Z}^n$$

Example: $f = x^n + 1$:

$$\begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-2} \\ c_{n-1} \end{bmatrix} = \begin{bmatrix} a_0 & -a_{n-1} & \dots & -a_1 \\ a_1 & a_0 & \dots & -a_2 \\ \vdots & & \ddots & \\ a_{n-2} & a_{n-3} & \dots & -a_{n-1} \\ a_{n-1} & a_{n-2} & \dots & a_0 \end{bmatrix} \cdot \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_{n-2} \\ b_{n-1} \end{bmatrix}$$

$\underbrace{\hspace{1.5cm}}_a \quad \underbrace{\hspace{1.5cm}}_{x \cdot a} \quad \underbrace{\hspace{1.5cm}}_{x^{n-1} \cdot a}$

Polynomial-LWE, SSTX'09

Let $f \in \mathbb{Z}[x]$ be unitary irreducible of degree n , $q \geq 2$

$$a = \sum_i a_i x^i \in \mathbb{Z}[x]/f \longrightarrow (a_0, \dots, a_{n-1}) \in \mathbb{Z}^n$$

Search Poly-LWE _{f} :

- ▶ Sample $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_q[x]/f$
- ▶ Sample a_i 's $\xleftarrow{\$} \mathbb{Z}_q[x]/f$
- ▶ Sample coefficients of $\mathbf{e}_i$ from χ

Given $(a_1, \dots, a_m)$ and

$(a_1 \cdot \mathbf{s} + \mathbf{e}_1, \dots, a_m \cdot \mathbf{s} + \mathbf{e}_m)$, find $\mathbf{s}$.

Polynomial-LWE, SSTX'09

Let $f \in \mathbb{Z}[x]$ be unitary irreducible of degree n , $q \geq 2$

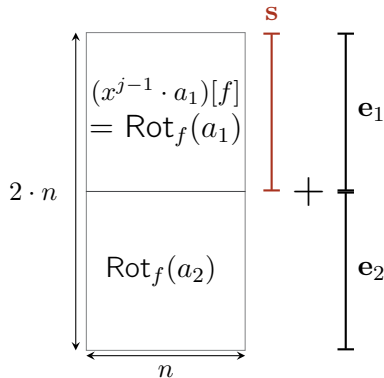
$$a = \sum_i a_i x^i \in \mathbb{Z}[x]/f \longrightarrow (a_0, \dots, a_{n-1}) \in \mathbb{Z}^n$$

Search Poly-LWE _{f} :

- ▶ Sample $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_q[x]/f$
- ▶ Sample a_i 's $\xleftarrow{\$} \mathbb{Z}_q[x]/f$
- ▶ Sample coefficients of $\mathbf{e}_i$ from χ

Given $(a_1, \dots, a_m)$ and

$(a_1 \cdot \mathbf{s} + \mathbf{e}_1, \dots, a_m \cdot \mathbf{s} + \mathbf{e}_m)$, find $\mathbf{s}$.



Polynomial-LWE, SSTX'09

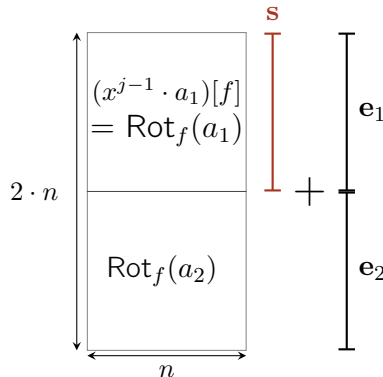
Let $f \in \mathbb{Z}[x]$ be unitary irreducible of degree n , $q \geq 2$

$$a = \sum_i a_i x^i \in \mathbb{Z}[x]/f \longrightarrow (a_0, \dots, a_{n-1}) \in \mathbb{Z}^n$$

Search Poly-LWE _{f} :

- ▶ Sample $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_q[x]/f$
- ▶ Sample a_i 's $\xleftarrow{\$} \mathbb{Z}_q[x]/f$
- ▶ Sample coefficients of $\mathbf{e}_i$ from χ

Given $(a_1, \dots, a_m)$ and
 $(a_1 \cdot \mathbf{s} + \mathbf{e}_1, \dots, a_m \cdot \mathbf{s} + \mathbf{e}_m)$, find $\mathbf{s}$.



One Poly-LWE samples $(a_i, a_i s + e_i)$ gives n 'standard' LWE samples

Can multiply two polynomials in time $\mathcal{O}(n \log q)$

Canonical embedding

Any polynomial can be represented as a vector via **canonical embedding**.

Let $f = x^n + 1$ be a polynomial with $n = 2^k$ (such polynomials are called cyclotomic).

Let $\omega_1, \dots, \omega_n \in \mathbb{C}$ be the roots of f : $\omega_i = e^{\frac{\pi i}{n}}$

Canonical embedding σ of a evaluates a in all ω_i :

$$\sigma : \sum_i a_i x^i \in \mathbb{Z}[x]/f \longrightarrow (a(\omega_0), \dots, a(\omega_{n-1})) \in \mathbb{C}^n$$

Canonical and coefficient embeddings are related via the Vandermonde matrix of ω_i 's:

$$V = \begin{bmatrix} 1 & \omega_1 & \dots & \omega_1^{n-1} \\ 1 & \omega_2 & \dots & \omega_2^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega_n & \dots & \omega_n^{n-1} \end{bmatrix}$$

$$(a(\omega_0), \dots, a(\omega_{n-1}))^T = V \cdot (a_0, \dots, a_{n-1})^T$$

Ring-LWE for $f = x^{2^k} + 1$, LPR'10

For $f = x^n + 1$, $n = 2^k$, $q \geq 2$, and

$\omega_1, \dots, \omega_n \in \mathbb{C}$ – the roots of f , V_f – Vandermonde matrix of ω_i 's

$$\sigma : \sum_i a_i x^i \in \mathbb{Z}[x]/f \longrightarrow (a(\omega_0), \dots, a(\omega_{n-1})) \in \mathbb{C}^n$$

Search Ring-LWE _{f} :

- ▶ Sample $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_q[x]/f$
- ▶ Sample a_i 's $\xleftarrow{\$} \mathbb{Z}_q[x]/f$
- ▶ Sample $\sigma(e_i)$'s from χ

Ring-LWE for $f = x^{2^k} + 1$, LPR'10

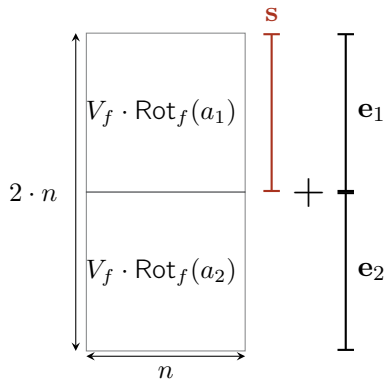
For $f = x^n + 1$, $n = 2^k$, $q \geq 2$, and

$\omega_1, \dots, \omega_n \in \mathbb{C}$ – the roots of f , V_f – Vandermonde matrix of ω_i 's

$$\sigma : \sum_i a_i x^i \in \mathbb{Z}[x]/f \longrightarrow (a(\omega_0), \dots, a(\omega_{n-1})) \in \mathbb{C}^n$$

Search Ring-LWE $_f$:

- ▶ Sample $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_q[x]/f$
- ▶ Sample a_i 's $\xleftarrow{\$} \mathbb{Z}_q[x]/f$
- ▶ Sample $\sigma(e_i)$'s from χ



Ring-LWE for $f = x^{2^k} + 1$, LPR'10

For $f = x^n + 1$, $n = 2^k$, $q \geq 2$, and

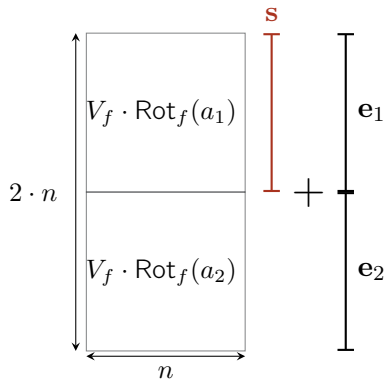
$\omega_1, \dots, \omega_n \in \mathbb{C}$ – the roots of f , V_f – Vandermonde matrix of ω_i 's

$$\sigma : \sum_i a_i x^i \in \mathbb{Z}[x]/f \longrightarrow (a(\omega_0), \dots, a(\omega_{n-1})) \in \mathbb{C}^n$$

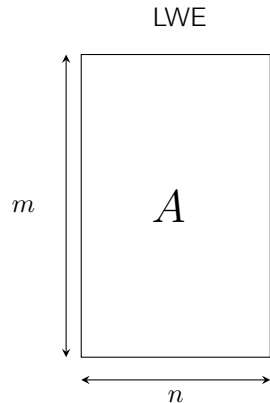
Search Ring-LWE _{f} :

- ▶ Sample $\mathbf{s} \xleftarrow{\$} \mathbb{Z}_q[x]/f$
- ▶ Sample a_i 's $\xleftarrow{\$} \mathbb{Z}_q[x]/f$
- ▶ Sample $\sigma(e_i)$'s from χ

- ▶ Can multiple in time $O(n \log q)$
- ▶ Poly-LWE and Ring-LWE are related for f 's with V_f of small operator norm



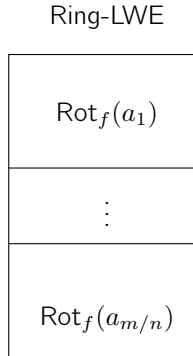
Module LWE



$$\mathbf{a}_i \in \mathbb{Z}_q^n$$

$$(\mathbf{a}_i, \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i)$$

$$\mathbf{s} \in \mathbb{Z}_q^n, e_i \in \mathbb{Z}$$

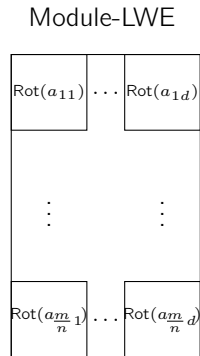


$$f = x^n + 1$$

$$a_i \in \mathbb{Z}[x]/f$$

$$(a_i, a_i \cdot \mathbf{s} + e_i)$$

$$\mathbf{s} \in \mathbb{Z}[x]/f, e_i \in \mathbb{Z}[x]/f$$



$$f = x^{n/d} + 1$$

$$\mathbf{a}_i \in (\mathbb{Z}[x]/f)^d$$

$$(\mathbf{a}_i, \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i)$$

$$\mathbf{s} \in (\mathbb{Z}[x]/f)^d, e_i \in \mathbb{Z}[x]/f$$

Kyber (ML-KEM) design

Alice



$$\text{pk} = (A \in \mathbb{Z}_q^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathbb{Z}_q^m)$$
$$\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$$

Decryption:

$$\text{if } |c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle| \text{ close to } 0 \implies \mu = 0$$

Recall PKE from LWE

Bob



$$\mu \in \{0, 1\}$$

Encryption:

$$\mathbf{t}_1 \leftarrow \chi^m, \mathbf{t}_2 \leftarrow \chi^n, \mathbf{t}_3 \leftarrow \chi^m$$

$$\mathbf{c}_1 = \mathbf{t}_1^T \cdot A + \mathbf{t}_2^T \bmod q$$

$$\xleftarrow{c = (\mathbf{c}_1, c_2)} \quad c_2 = \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 + \mu \lfloor \frac{q}{2} \rfloor \bmod q$$

Kyber (ML-KEM) design

Alice



Recall PKE from LWE

Change $\mathbb{Z}_q$ to $\mathbb{Z}_q[x]/(x^{256} + 1) := \mathcal{R}$

$$\text{pk} = (A \in \mathcal{R}^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathcal{R}^m)$$
$$\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$$

Decryption:

$$\text{if } |c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle| \text{ close to } 0 \implies \mu = 0$$

Bob



$$\mu \in \{0, 1\}$$

Encryption:

$$\mathbf{t}_1 \leftarrow \chi^m, \mathbf{t}_2 \leftarrow \chi^n, \mathbf{t}_3 \leftarrow \chi^m$$

$$\mathbf{c}_1 = \mathbf{t}_1^T \cdot A + \mathbf{t}_2^T \bmod q$$

$$\xleftarrow{c = (\mathbf{c}_1, c_2)} \quad c_2 = \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 + \mu \lfloor \frac{q}{2} \rfloor \bmod q$$

Kyber (ML-KEM) design

Alice



Recall PKE from LWE

Change $\mathbb{Z}_q$ to $\mathbb{Z}_q[x]/(x^{256} + 1) := \mathcal{R}$

Now Bob encrypts a vector

χ is a distribution over $\mathcal{R}$

$$\text{pk} = (A \in \mathcal{R}^{m \times n}, \mathbf{b} = A\mathbf{s} + \mathbf{e} \in \mathcal{R}^m)$$

$$\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$$

Decryption (coefficient-wise):

$$\text{if } |c_2 - \langle \mathbf{c}_1, \mathbf{s} \rangle| \text{ close to } 0 \implies \mu = 0$$

Bob



$$\mu \in \{0, 1\}^n$$

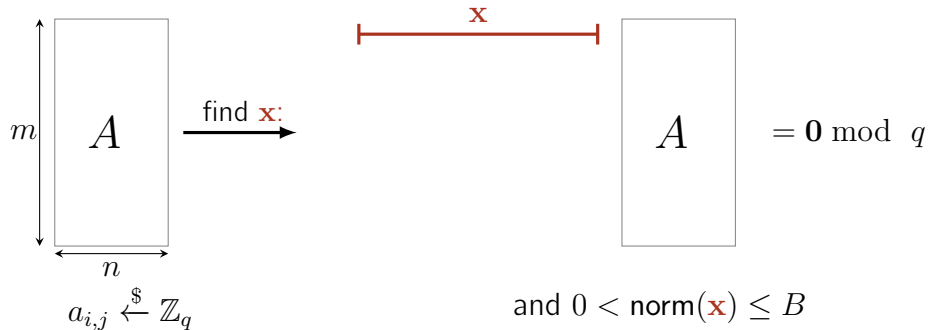
Encryption:

$$\mathbf{t}_1 \leftarrow \chi^m, \mathbf{t}_2 \leftarrow \chi^n, \mathbf{t}_3 \leftarrow \chi^m$$

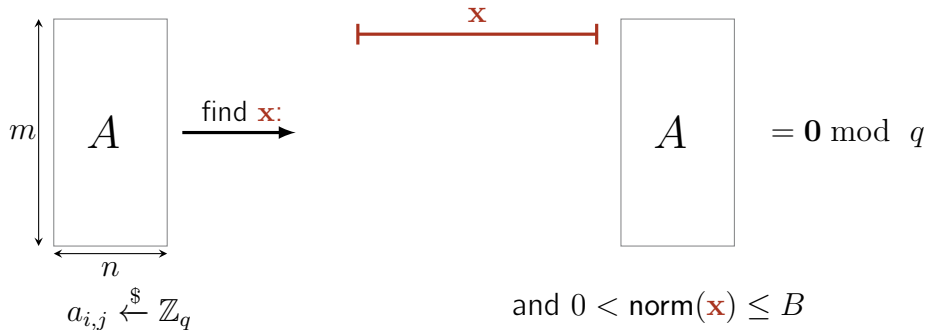
$$\mathbf{c}_1 = \mathbf{t}_1^T \cdot A + \mathbf{t}_2^T \bmod q$$

$$\xleftarrow{c = (\mathbf{c}_1, c_2)} \quad c_2 = \langle \mathbf{t}_1, \mathbf{b} \rangle + \mathbf{t}_3 + \mu \lfloor \frac{q}{2} \rfloor \bmod q$$

Short Integer Solution (SIS), Ajtai'96



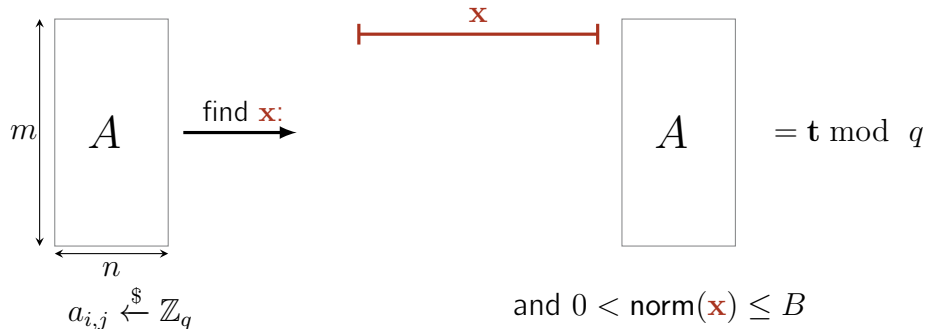
Short Integer Solution (SIS), Ajtai'96



The problem is interesting in the Euclidean norm: $\|\mathbf{x}\| \leq B$

and in the infinity norm: $\|\mathbf{x}\|_{\infty} = \max_i \{|\mathbf{x}_i|\} \leq B$

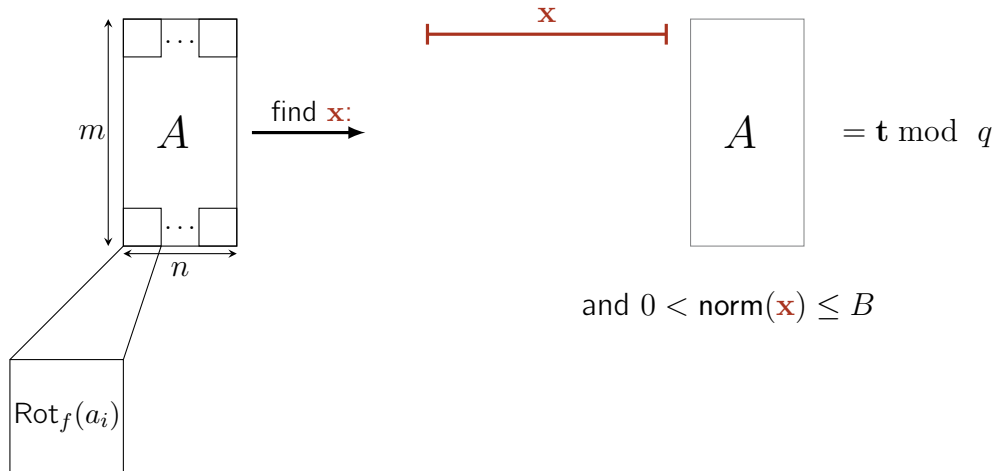
Inhomogenous Short Integer Solution (ISIS), Ajtai'96



The problem is interesting in the Euclidean norm: $\|\mathbf{x}\| \leq B$

and in the infinity norm: $\|\mathbf{x}\|_{\infty} = \max_i \{|\mathbf{x}_i|\} \leq B$

Module SIS over $\mathbb{Z}[x]/f$



(Module)-SIS hardness

- ▶ Inhomogenous version and homogenous versions are equivalent (think why)
- ▶ The larger B – the upper bound on the solution, the easier the problem is ($B = q$ is trivial)
- ▶ For very small B (and m) a solution may not exist
- ▶ Typically, B is chosen such that there are exponentially many solutions ($B^m > q^n$)
- ▶ The larger n , the harder the problem is
- ▶ Interesting parameters to solve will be $n = 1024$, $B = q/8$

Methods to construct a signature scheme

I. Hash-and-Sign Paradigm

Examples: Falcon = NTRUSign + [GPV08]

Pros: short signatures (0.7kB for Falcon 512), fast verification

Cons: complex to implement, expensive KeyGen

II. Fiat-Shamir Heuristic [FS]

Examples: Bai-Gabraith, Dilithium, HAWK (not based on SIS)

Pros: easy to implement, fast KeyGen, Sign

Cons: larger signatures (2.7kB for Dilithium3)

Dilithium simplified

χ – a distribution over $R = \mathbb{Z}_q[x]/(x^{256+1})$ that outputs ‘short’ polynomials

$\mathcal{H}$ – a cryptographic hash function that outputs short digests

1. KeyGen :

1. $A \leftarrow R_q^{k \times \ell}$
2. $\mathbf{s} \leftarrow \chi^k$
3. $\mathbf{t} = \mathbf{s}A$
4. $\text{vk} = (A, \mathbf{t}), \text{sk} = (\mathbf{s}, A, \mathbf{t})$.

Dilithium simplified

χ – a distribution over $R = \mathbb{Z}_q[x]/(x^{256+1})$ that outputs ‘short’ polynomials

$\mathcal{H}$ – a cryptographic hash function that outputs short digests

I. KeyGen :

1. $A \leftarrow R_q^{k \times \ell}$
2. $\mathbf{s} \leftarrow \chi^k$
3. $\mathbf{t} = \mathbf{s}A$
4. $\text{vk} = (A, \mathbf{t}), \text{sk} = (\mathbf{s}, A, \mathbf{t}).$

II. Sign(sk, m) :

1. $\mathbf{r} \leftarrow \chi^\ell$
2. $\mathbf{u} = \mathbf{r}A$
3. $c = \mathcal{H}(m || \mathbf{u})$
4. $\mathbf{z} = \mathbf{r} + c\mathbf{s}$
5. **return** $\sigma = (\mathbf{u}, \mathbf{z})$

Dilithium simplified

χ – a distribution over $R = \mathbb{Z}_q[x]/(x^{256+1})$ that outputs ‘short’ polynomials

$\mathcal{H}$ – a cryptographic hash function that outputs short digests

I. KeyGen :

1. $A \leftarrow R_q^{k \times \ell}$
2. $\mathbf{s} \leftarrow \chi^k$
3. $\mathbf{t} = \mathbf{s}A$
4. $\text{vk} = (A, \mathbf{t}), \text{sk} = (\mathbf{s}, A, \mathbf{t}).$

II. Sign(sk, m) :

1. $\mathbf{r} \leftarrow \chi^\ell$
2. $\mathbf{u} = \mathbf{r}A$
3. $c = \mathcal{H}(m || \mathbf{u})$
4. $\mathbf{z} = \mathbf{r} + c\mathbf{s}$
5. **return** $\sigma = (\mathbf{u}, \mathbf{z})$

III. Verify(vk, m, $\sigma = (\mathbf{u}, \mathbf{z})$) :

1. $c = \mathcal{H}(m || \mathbf{u})$
2. If $\mathbf{z}$ is short and $\mathbf{z}A - c\mathbf{t} == \mathbf{u}$:
 return “Accept”
Else **return** “Reject”

Dilithium simplified

χ – a distribution over $R = \mathbb{Z}_q[x]/(x^{256+1})$ that outputs ‘short’ polynomials

$\mathcal{H}$ – a cryptographic hash function that outputs short digests

I. KeyGen :

1. $A \leftarrow R_q^{k \times \ell}$
2. $\mathbf{s} \leftarrow \chi^k$
3. $\mathbf{t} = \mathbf{s}A$ Module SIS
4. $\text{vk} = (A, \mathbf{t}), \text{sk} = (\mathbf{s}, A, \mathbf{t}).$

II. Sign(sk, m) :

1. $\mathbf{r} \leftarrow \chi^\ell$
2. $\mathbf{u} = \mathbf{r}A$ Module SIS
3. $c = \mathcal{H}(m || \mathbf{u})$
4. $\mathbf{z} = \mathbf{r} + c\mathbf{s}$
5. **return** $\sigma = (\mathbf{u}, \mathbf{z})$

III. Verify(vk, m, $\sigma = (\mathbf{u}, \mathbf{z})$) :

1. $c = \mathcal{H}(m || \mathbf{u})$
2. If $\mathbf{z}$ is short and $\mathbf{z}A - c\mathbf{t} == \mathbf{u}$:
 return “Accept”
 Else **return** “Reject”

Correctness:

- ▶ $\mathbf{z}A - c\mathbf{t} = \mathbf{r}A + c\mathbf{s} \cdot A - c\mathbf{s} \cdot A = \mathbf{r}A$
- ▶ $\mathbf{z}$ short since $\mathbf{r}, c, \mathbf{s}$ are short

Dilithium simplified

χ – a distribution over $R = \mathbb{Z}_q[x]/(x^{256+1})$ that outputs ‘short’ polynomials

$\mathcal{H}$ – a cryptographic hash function that outputs short digests

I. KeyGen :

1. $A \leftarrow R_q^{k \times \ell}$
2. $\mathbf{s} \leftarrow \chi^k$
3. $\mathbf{t} = \mathbf{s}A$ Module SIS
4. $\text{vk} = (A, \mathbf{t}), \text{sk} = (\mathbf{s}, A, \mathbf{t}).$

II. Sign(sk, m) :

1. $\mathbf{r} \leftarrow \chi^\ell$
2. $\mathbf{u} = \mathbf{r}A$ Module SIS
3. $c = \mathcal{H}(m || \mathbf{u})$
4. $\mathbf{z} = \mathbf{r} + c\mathbf{s}$ Leaks $\mathbf{s}$!
5. **return** $\sigma = (\mathbf{u}, \mathbf{z})$

III. Verify(vk, m, $\sigma = (\mathbf{u}, \mathbf{z})$) :

1. $c = \mathcal{H}(m || \mathbf{u})$
2. If $\mathbf{z}$ is short and $\mathbf{z}A - c\mathbf{t} == \mathbf{u}$:
 return “Accept”
 Else **return** “Reject”

Correctness:

- ▶ $\mathbf{z}A - c\mathbf{t} = \mathbf{r}A + c\mathbf{s} \cdot A - c\mathbf{s} \cdot A = \mathbf{r}A$
- ▶ $\mathbf{z}$ short since $\mathbf{r}, c, \mathbf{s}$ are short

Dilithium simplified

χ – a distribution over $R = \mathbb{Z}_q[x]/(x^{256+1})$ that outputs ‘short’ polynomials

$\mathcal{H}$ – a cryptographic hash function that outputs short digests

I. KeyGen :

1. $A \leftarrow R_q^{k \times \ell}$
2. $\mathbf{s} \leftarrow \chi^k$
3. $\mathbf{t} = \mathbf{s}A$ Module SIS
4. $\text{vk} = (A, \mathbf{t}), \text{sk} = (\mathbf{s}, A, \mathbf{t}).$

II. Sign(sk, m) :

1. $\mathbf{r} \leftarrow \chi^\ell$
2. $\mathbf{u} = \mathbf{r}A$ Module SIS
3. $c = \mathcal{H}(m || \mathbf{u})$
4. $\mathbf{z} = \mathbf{r} + c\mathbf{s}$ Leaks $\mathbf{s}$! Solution: hide the distribution of $\mathbf{z}$ by rejecting some $\mathbf{z}$
5. **return** $\sigma = (\mathbf{u}, \mathbf{z})$

III. Verify(vk, m, $\sigma = (\mathbf{u}, \mathbf{z})$) :

1. $c = \mathcal{H}(m || \mathbf{u})$
2. If $\mathbf{z}$ is short and $\mathbf{z}A - c\mathbf{t} == \mathbf{u}$:
 return “Accept”
 Else **return** “Reject”

Correctness:

- ▶ $\mathbf{z}A - c\mathbf{t} = \mathbf{r}A + c\mathbf{s} \cdot A - c\mathbf{s} \cdot A = \mathbf{r}A$
- ▶ $\mathbf{z}$ short since $\mathbf{r}, c, \mathbf{s}$ are short

Beyond PKE and Signature



Useful resources



References

content...